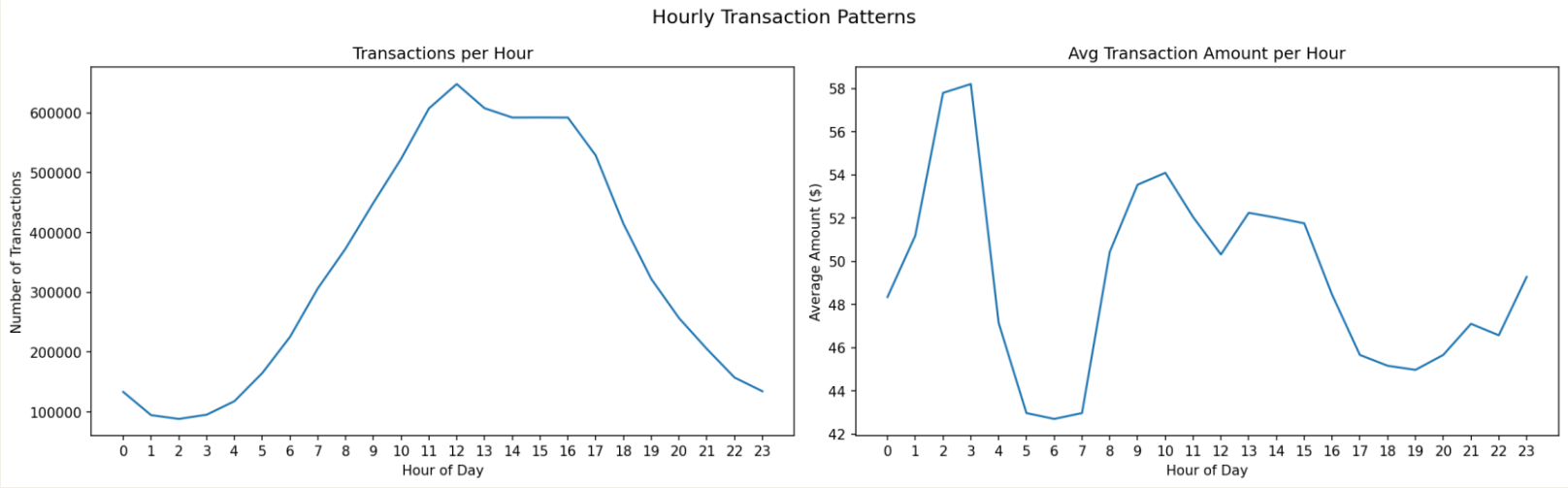
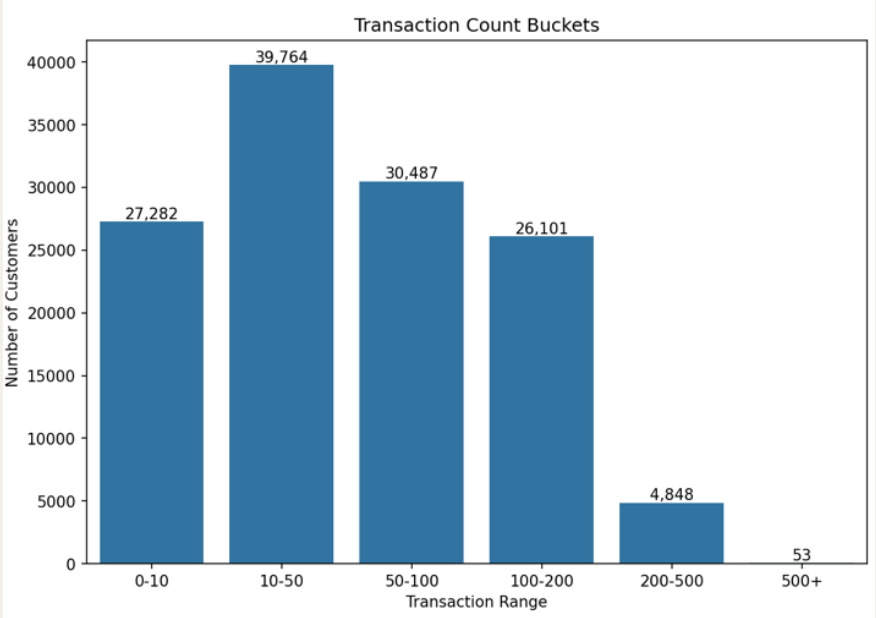


FinSights

DS 5500 : Check-in 03

Omkar Sonar

23rd February 2026



Quick Refresher

- Total Customers → 68,857
- Transactions → 8.2 M
- Observation window → 65 days
- Features built so far → ~40

Primary Business Use Cases

1. Identifying good candidates for the bank's rewards program
2. Personalized cashback incentives
3. Promote local businesses

Merchant Segmentation via MCC Codes

What are MCC Codes?
4-digit identifiers assigned to every merchant describing the type of business

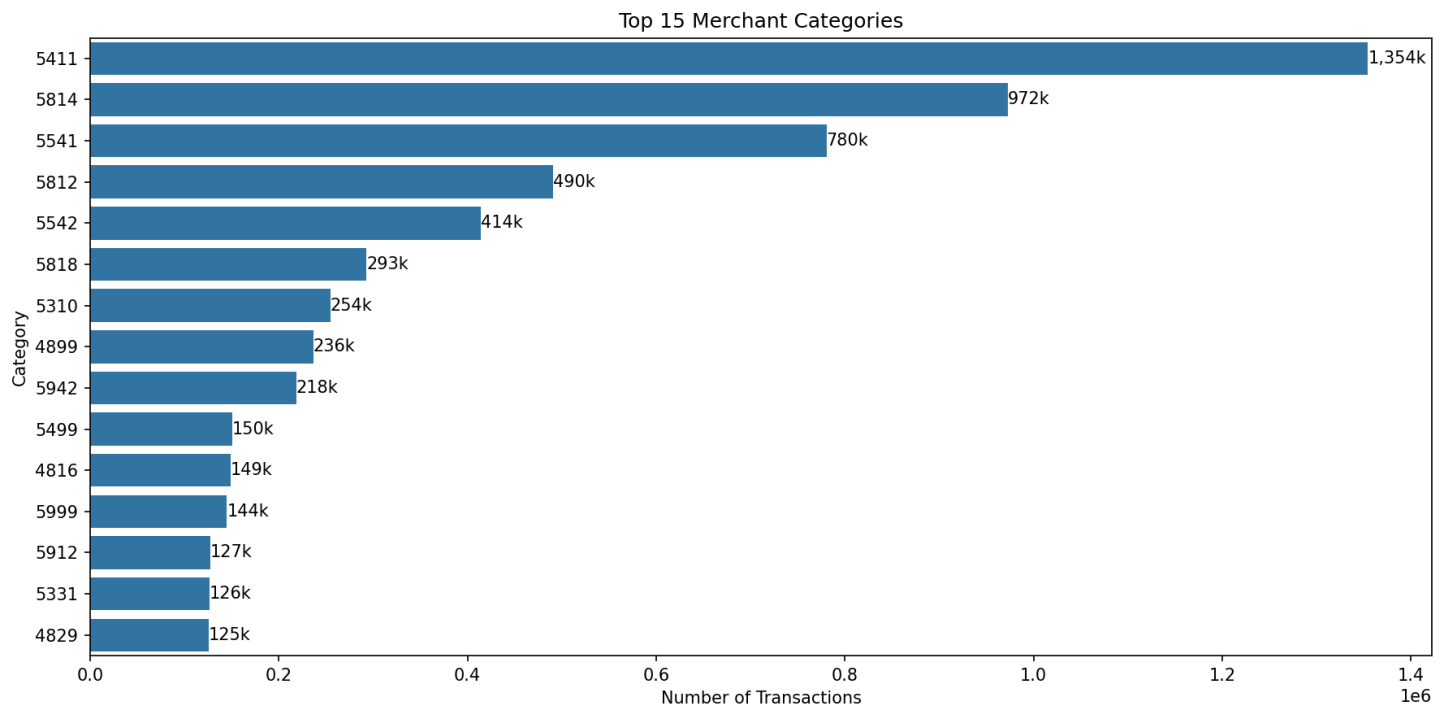
Grocery → 5411,5499,5422,5441,5451,5462

Dining → 5812,5814,5813,5811

Gas → 5541,5542,5983

Retail → 5310,5311,5331,5399,5300,5999

....



Feature Engineering

This was done on the basis of temporal features (when customers spend), merchant categories, customer demographics, spending trends. Not all features will be useful but this will help us in identifying the customer segments.

	0
CustomerID	BSB008064
total_spend	4363.27
transaction_count	83
avg_transaction_amount	52.569518
median_transaction_amount	22.22
std_transaction_amount	72.682751
max_transaction_amount	398.36
min_transaction_amount	2.99
grocery_share	0.354697
dining_share	0.0
gas_share	0.099148
retail_share	0.211743
digital_share	0.006112
utilities_share	0.068504
healthcare_share	0.0
home_share	0.0
financial_share	0.100842
transport_share	0.0
entertainment_share	0.146871
clothing_share	0.012083
other_share	0.0
weekend_share	0.409639
avg_transactions_per_week	13.833333
dec_to_nov_daily_spend_ratio	1.547932
unique_merchants	40
top_merchant_share	0.295886
repeat_merchant_rate	0.175
repeat_transaction_rate	0.518072
avg_visits_per_merchant	2.075

Age	37.0
RelationshipYears	4.0
is_male	0
is_female	1
NumberActiveDDAs	2.0
NumberActiveTimeDeposits	0.0
NumberActiveLoans	1.0
NumberCreditCardAccts	0.0
has_deposit	1
has_loan	1
has_creditcard	0
has_atm	0
has_debit	1
is_primary	1
is_wealth	0
has_payroll	0
dec_to_nov_avg_amount_ratio	1.327566

Insights from Stakeholder

- Current direction of the project was appreciated
- Discussion on grouping with ATM_Address and ATM_City_State (find out local merchants)